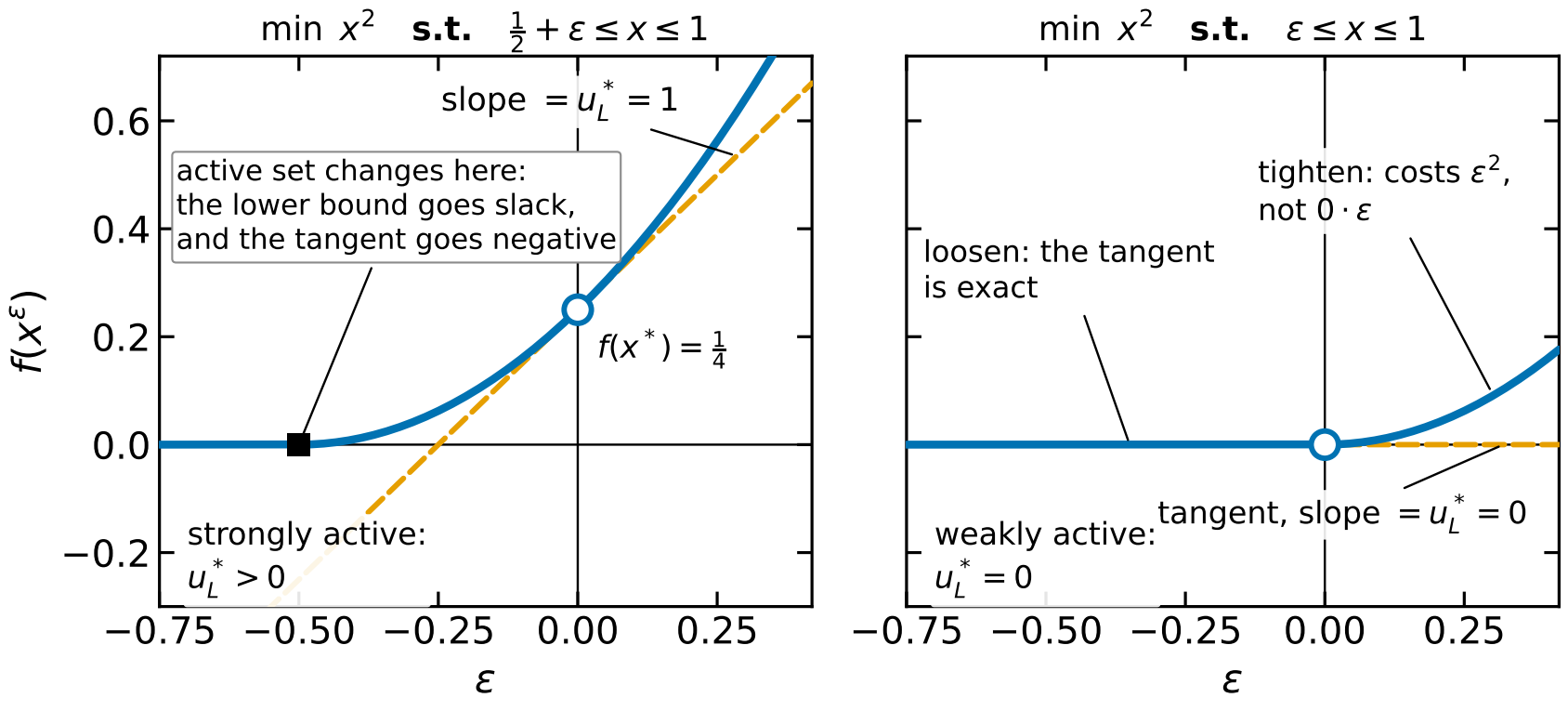
min x^2 **s.t.** $\frac{1}{2} + \varepsilon \le x \le 1$
slope $= u_L^* = 1$
active set changes here:
the lower bound goes slack,
and the tangent goes negative
$f(x^*) = \frac{1}{4}$
strongly active:
$u_L^* > 0$
$f(x^\varepsilon)$
0.6
0.4
0.2
0.0
−0.2
−0.75
−0.50
−0.25
0.00
0.25
ε
min x^2 **s.t.** $\varepsilon \le x \le 1$
tighten: costs ε^2,
not $0 \cdot \varepsilon$
loosen: the tangent
is exact
tangent, slope $= u_L^* = 0$
weakly active:
$u_L^* = 0$
−0.75
−0.50
−0.25
0.00
0.25
ε